

1. Administrative & Study Identification

Full Study Title: [A Phase 3, Randomized, Open-label, Study to Compare the Efficacy and Safety of Adjuvant MK-2870 in Combination with Pembrolizumab (MK-3475) Versus Treatment of Physician's Choice (TPC) in Participants With Triple Negative Breast Cancer (TNBC) Who Received Neoadjuvant Therapy and Did Not Achieve a Pathological Complete Response (pCR) at Surgery] **Short Title/Acronym:** [Sacituzumab Tirumotecan (MK-2870) Plus Pembrolizumab Versus TPC in TNBC Who Did Not Achieve pCR (MK-2870-012)] **Protocol Number:** [MK-2870-012] **NCT Number:** [NCT06393374] **Posting ID:** [1075503203919023793] **Trial Status:** [RECRUITING - Currently recruiting participants] **Sponsor / Company Name:** [Merck Sharp & Dohme LLC / Merck] **Investigational Product:** [Sacituzumab tirumotecan (MK-2870 / sac-TMT) and Pembrolizumab; Capecitabine as part of TPC] **Phase of Development:** [Phase 3] **Medical Monitor:** [Sponsor Clinical Director / Merck Sharp & Dohme LLC]

2. Study Synopsis & Rationale

2.1 Study Objective * Primary Objective: [To compare adjuvant sacituzumab tirumotecan plus pembrolizumab to treatment of physician's choice (TPC: pembrolizumab or pembrolizumab plus capecitabine) with respect to invasive disease-free survival (iDFS) per investigator assessment. It is hypothesized that the investigational combination is superior to TPC.] *** Secondary Objectives:** [To evaluate overall survival (OS), distant recurrence-free survival (DRFS), changes in health-related quality of life, overall safety and tolerability profile, and to assess efficacy across subgroups defined by TROP2 expression.]

2.2 Background & Rationale [Trophoblast cell surface antigen 2 (TROP2) expression is typically higher in triple-negative breast cancer (TNBC) than in other breast cancer subtypes, which is associated with poor prognosis. Sacituzumab tirumotecan (MK-2870) is a novel antibody-drug conjugate (ADC) consisting of an anti-TROP2 antibody coupled to a cytotoxic belotecan derivative. The current standard of care for patients with newly diagnosed high-risk early-stage TNBC is the KEYNOTE-522 regimen (neoadjuvant pembrolizumab plus chemotherapy followed by adjuvant pembrolizumab). This study addresses the medical necessity to improve outcomes for TNBC patients who undergo standard neoadjuvant therapy but fail to achieve a pathological complete response (pCR) at surgery, by investigating whether adding the TROP2 ADC MK-2870 to adjuvant pembrolizumab provides superior iDFS compared to TPC.]

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** [Interventional] * **Control Type:** [Active Comparator (Treatment of Physician's choice - TPC)] * **Blinding:** [Open-label] * **Randomization:** [Randomized (Parallel Assignment)]

3.2 Planned Enrollment & Duration * **Target N\$:** [Approximately 1,530 evaluable patients] * **Number of Sites:** [Global multicenter study] * **Estimated Study Start:** [24-Jun-2024] * **Estimated Primary Completion:** [16-Dec-2030]

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. [Age $\geq$ 18 years with centrally confirmed Triple-Negative Breast Cancer (TNBC), as defined by the most recent ASCO/CAP guidelines]. 2. [Received prior neoadjuvant treatment based on the KEYNOTE-522 regimen (pembrolizumab with carboplatin/taxanes and pembrolizumab with anthracycline-based chemotherapy) followed by surgery according to NCCN guidelines]. 3. [Has non-pathologic complete response (non-pCR) at surgery, with adequate excision of all clinically evident disease and adequate recovery from surgery]. 4. [No evidence of locoregional or distant relapse as assessed by the treating physician]. 5. [Provided tissue from the surgical resection for central laboratory determination of TROP2 status]. 6. [Randomization conducted within 16 weeks from surgical resection, and completed/recovered from adjuvant radiation therapy (if indicated)]. 7. [ECOG performance status of 0 to 1 assessed within 7 days before first dose]. 8. [Recovered to $\leq$ Grade 1 or baseline from prior anticancer therapy AEs (except alopecia)]. 9. [HIV-infected participants must have well-controlled HIV on antiretroviral therapy (ART). HBsAg positive participants are eligible if on HBV antiviral therapy for $\geq$ 4 weeks with undetectable viral load]. 10. [Strict adherence to contraception and sperm donation restrictions for a specified period after study intervention, and females must not be pregnant or breastfeeding].

4.2 Exclusion Criteria 1. [Known germline BRCA mutation (deleterious/suspected) eligible for adjuvant therapy with olaparib where approved/available]. 2. [Grade >2 peripheral neuropathy]. 3. [History of documented severe dry eye syndrome, severe Meibomian gland disease, blepharitis, or severe corneal disease]. 4. [Active inflammatory bowel disease (Crohn's, ulcerative colitis) requiring immunosuppressants, or chronic diarrhea]. 5. [Uncontrolled significant cardiovascular/cerebrovascular disease (e.g., NYHA Class III/IV heart failure, unstable angina, MI, QTcF prolongation >480 ms) within 6 months]. 6. [Prior treatment with a TROP2-directed ADC or topoisomerase I inhibitor-containing ADC]. 7. [Received anticancer therapy in the adjuvant phase (chemo, small molecules, PARP inhibitors,

immunotherapies, other ADCs), excluding adjuvant radiation]. 8. [Current use of strong CYP3A4 inducer/inhibitor requiring a 2-week washout before MK-2870]. 9. [Prior therapy with anti-PD-1, anti-PD-L1/L2, CTLA-4, OX-40, or CD137 (except for pembrolizumab as neoadjuvant therapy)]. 10. [Received live/live-attenuated vaccine within 30 days before the first dose]. 11. [History of noninfectious pneumonitis/interstitial lung disease (ILD) requiring steroids, or current pneumonitis/ILD]. 12. [Diagnosis of immunodeficiency or receiving chronic systemic steroids (>10 mg daily prednisone equivalent) within 7 days prior to first dose]. 13. [Active autoimmune disease requiring systemic treatment past 2 years (replacement therapy allowed)]. 14. [Additional progressing malignancy (active treatment past 5 years), active infection requiring systemic therapy, concurrent active HBV and HCV, HIV with Kaposi's sarcoma/Multicentric Castleman's, or history of allogeneic tissue/solid organ transplant].

5. Intervention & Treatment Plan

5.1 Investigational Regimen * Dosage/Frequency: * **Experimental Arm:** [Sacituzumab tirumotecan (MK-2870) administered every 2 weeks + Pembrolizumab administered every 6 weeks]. * **Active Comparator Arm (TPC):** [Pembrolizumab every 6 weeks alone, OR Pembrolizumab + Capecitabine (chemotherapy taken orally twice a day for 2 out of every 3 weeks)]. * **Route of Administration:** [Intravenous (IV) for MK-2870 and Pembrolizumab; Oral for Capecitabine]. * **Duration of Treatment:** [Treatment continues for about 6 months, unless disease progression occurs or treatment is not tolerated].

5.2 Concomitant & Prohibited Therapy * Permitted: [Adjuvant radiation therapy (must be completed and recovered before randomization), approved replacement therapies]. * **Prohibited:** [Other concurrent investigational agents, live vaccines, strong CYP3A4 inducers/inhibitors, or systemic anti-cancer therapies outside the protocol-defined regimens].

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	[Up to 16 weeks post-surgery]	[Informed Consent, Medical History, TROP2 central testing, Imaging, Eligibility Assessment]
Baseline	Day 0	[Randomization, First Dose, EORTC QLQ-C30 baseline assessment]
Interim	Weeks [2-6] cycle	[Dosing IV/Oral, Safety Labs, Tumor Imaging, Physical Exams, ECG, QLQ Questionnaires]
End of Study	Follow-up	[Long-term survival and recurrence follow-up for up to 10 years]

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** [Invasive disease-free survival (iDFS) - defined as the time from randomization to invasive local, regional, or distant recurrence, invasive contralateral breast cancer, or death due to any cause]. * **Measurement Method:** [Investigator assessment].

7.2 Secondary Endpoints * [Overall Survival (OS)]. * [Distant recurrence-free survival (DRFS)]. * [Change from baseline in European Organization for the Research and Treatment of Cancer Quality of Life Questionnaire Core 30 (EORTC QLQ-C30) Global Health Status (Item 29) and Quality of Life (Item 30) Combined Score]. * [Overall safety and tolerability profile (Adverse Events)]. * [Efficacy based on central TROP2 expression levels].

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** [Clinical evaluations, regular laboratory tests, physical examinations, and ECGs during treatment cycles].
 - **Serious Adverse Events (SAEs):** [Standard clinical trial safety reporting to the Sponsor and regulatory authorities].
 - **Data Monitoring Committee (DMC):** [Independent Data Monitoring Committee to routinely review unblinded safety and efficacy data].
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** [Intent-to-Treat (ITT) population for all primary efficacy analyses; Safety Set for all patients receiving at least one dose of study drug].
 - **Sample Size Justification:** [Targeting ~1,530 participants to be adequately powered to test the superiority hypothesis of MK-2870 + pembrolizumab over TPC for iDFS].
 - **Handling of Missing Data:** [Standard time-to-event censoring rules (e.g., Kaplan-Meier methodology) for survival outcomes].
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** [Regular onsite and remote monitoring of trial sites by the Sponsor (Merck) or assigned CRO].
 - **Audit Trail:** [Electronic Case Report Form (eCRF) utilizing 21 CFR Part 11 compliant data capture systems].
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11. Study Identifiers & Governance

- **Primary ID:** {{MK-2870-012}}
 - **Registry Number:** {{NCT06393374}}
 - **Posting ID:** {{1075503203919023793}}
 - **Sponsor/Lead Organization:** {{Merck Sharp & Dohme LLC / Merck}}
 - **Study Title:** {{Sacituzumab Tirumotecan (MK-2870) Plus Pembrolizumab Versus TPC in TNBC Who Did Not Achieve pCR}}
 - **Official Regulatory Title:** {{A Phase 3, Randomized, Open-label, Study to Compare the Efficacy and Safety of Adjuvant MK-2870 in Combination with Pembrolizumab (MK-3475) Versus Treatment of Physician's Choice (TPC) in Participants With Triple Negative Breast Cancer (TNBC) Who Received Neoadjuvant Therapy and Did Not Achieve a Pathological Complete Response (pCR) at Surgery}}
 - **Trial Status:** {{RECRUITING (Currently recruiting participants)}}
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12. Clinical Framework

- **Primary Condition / Disease Area:** {{Triple-Negative Breast Cancer (TNBC) Finding}}
 - **Condition Nomenclature:** {{MeSH Code: D063852 | SNOMED ID: 706970001}}
 - **Diagnosis Threshold:** {{Centrally confirmed TNBC with non-pCR at surgery following KEYNOTE-522 neoadjuvant regimen}}
 - **Medical Methodology:** {{Adjuvant targeted therapy (TROP2 ADC) plus Immunotherapy (Pembrolizumab) +/- Chemotherapy (Capecitabine - ATC: L01BC06)}}
 - **Optimization Target:** {{Improvement in invasive disease-free survival (iDFS) and overall survival (OS)}}
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13. Study Procedures & Methodology

- **Management Pathway:** {{Adjuvant clinical pathway for high-risk early-stage TNBC with residual disease}}
 - **Education Protocol (HCP):** {{Protocol-specific training, safety monitoring instruction for ADCs and immunotherapies}}
 - **Education Protocol (Patient):** {{Informed consent, orientation on study procedures, and dosing schedules}}
 - **Quality Audit Mechanism:** {{Routine sponsor monitoring, central laboratory assessment for TROP2}}
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14. Observation & Follow-Up Schedule

- **Total Duration:** {{Up to 10 years of follow-up}}
- **Onsite Visit Cadence:** {{Every 2 weeks for experimental arm IV infusions; up to every 6 weeks based on treatment cycle}}

